

A 4-Step Process for Effective System Design Interview

System design interviews are **open-ended**, meaning there is no single correct solution. The goal is to evaluate:

- Problem-solving approach
- System thinking
- Trade-offs
- Communication clarity

A strong candidate follows a **structured approach**, not a rushed solution.

Step 1: Understand the Problem & Define Scope

Don't Be Like "Jimmy"

Jimmy immediately answers without understanding requirements.

None

Jumping to a solution = Wrong approach in system design interviews

System design is NOT a trivia test.
There is no “instant correct answer”.

Correct Approach

Start by:

- Clarifying requirements
- Understanding constraints
- Defining scope clearly
- Asking intelligent questions

Why This Step Matters

If you don't define scope properly:

- You may design the wrong system
- You may miss key components
- Your solution becomes irrelevant

Good systems start with **good questions**

Types of Questions to Ask

1. Functional Requirements

None

- What features are required?
- What should the system do?

Example:

- Post a tweet
- View timeline
- Like/comment system

2. User Scale

None

- How many users?
- Daily active users (DAU)?
- Expected growth?

3. Traffic Expectations

None

- Read-heavy or write-heavy system?
- Peak traffic?

4. System Growth

None

- 3 months scale?
- 6 months scale?
- 1 year scale?

5. Data Types

None

- Text only?
- Images/videos?
- File uploads?

6. Tech Stack Constraints

None

- Any existing services?
- Preferred databases or cloud providers?

Example: News Feed System Discussion

Candidate vs Interviewer Flow

Step 1: Platform

Candidate:

Is this a mobile app or web app?

Interviewer:

Both

Step 2: Core Features

Candidate:

What are the most important features?

Interviewer:

Post content + view news feed

Step 3: Feed Ordering

Candidate:

Is feed chronological or ranked?

Interviewer:

Reverse chronological for simplicity

Step 4: Social Graph Size

Candidate:

How many friends per user?

Interviewer:

5000 friends

Step 5: Scale

Candidate:

Expected traffic?

Interviewer:

10 million DAU

Step 6: Media Support

Candidate:

Does post include media?

Interviewer:

Yes (images + videos)

Key Insight from This Step

None

Good system design starts with correct assumptions, not correct diagrams

Interview Strategy Tip

Always Do This First:

- Ask clarifying questions
- Write assumptions clearly
- Confirm scope with interviewer

Avoid:

- Jumping to architecture immediately
- Designing databases too early
- Skipping requirement gathering

Mental Model

None

Understand → Clarify → Constrain → Then Design

One-Line Summary

Step 1 in system design interviews is to fully understand the problem by asking clarifying questions, defining scope, and setting assumptions before jumping into design.